Auto-Decte: Trust-Constrained Document Intelligence through Candidate–Fact Separation and Selective Prediction

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Abstract

Document intelligence in payroll-adjacent and production workflows must control both recognition error and the authority granted to machine output. Auto-Decte is a trust-constrained human-in-the-loop architecture for structured industrial forms. Its central invariant separates immutable evidence, append-only machine candidates, attributable human decisions, and versioned facts: perception, retrieval, and language-model services have no direct fact-writing capability through the evaluated application interfaces. The implemented authority contract makes this invariant executable: every recognition result is persisted as a content-addressed candidate certificate bound to its field, evidence, and expected fact version; a confirmation is a per-field attributable human decision over a database-loaded certificate, committed with the record version, decisions, transitions, and audit event in a single compare-and-swap-protected transaction; and reverse traces rebuild each field chain from evidence through certificate, decision, and transition, reporting any tampered, missing, or wrongly bound entry as incomplete.

We evaluated the implementation at three levels. First, a seed-disjoint cell benchmark measured recognition: a HOG–linear-SVM recognizer achieved 99.81% accuracy at full coverage, and 60 synthetic forms under five whole-page conditions achieved 60% complete-form success, exposing QR and marker failures under blur and rotation. Second, an executable authority-integration benchmark ran the real services over a synthetic fault corpus with a scripted reviewer: every injected corruption (tampered transition producer, deleted

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transition, rebound record-version row) was detected as an incomplete trace, cross-field certificate substitution and cross-form evidence were rejected before any write, all valid certificate confirmations succeeded, and stale replay was contained. Third, the integrated software verification suite passes 239 automated tests, and the benchmark outputs are byte-identical across repeated runs under an immutable manifest. The evidence supports the executable authority boundary but not production effectiveness: independently transcribed real forms and controlled reviewer studies remain necessary.

Keywords: document intelligence, human-in-the-loop, selective prediction, data provenance, trustworthy artificial intelligence

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